Keshav Rajput

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Security software engineer, new grad. MS Computer Science, NYU Tandon, May 2026. Post-quantum cryptography, audits, fuzzing.
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EDUCATION

New York University, Tandon School of Engineering Sep 2024 - May 2026
Master of Science (MS) in Computer Science GPA 3.67/4.0

Indraprastha Institute of Information Technology (IIIT) Delhi 2020 - 2024
Bachelor of Technology (B.Tech) in Computer Science and Applied Mathematics, minor in Economics CGPA 8.19/10

Coursework: Applied Cryptography, Post-Quantum Cryptography, Computer Networks, Operating Systems, Distributed Systems, Design and Analysis of Algorithms II

PROJECTS

mldsa-auth | C11, liboqs, libsodium, SQLite, CMake | github.com/happyc0der/mldsa-auth 2026
- Built a post-quantum mutual authentication protocol in C11: ML-DSA-65 (FIPS 204) signatures over hybrid X25519 + ML-KEM-768, HKDF-SHA256, ChaCha20-Poly1305 records.
- Measured a 0.45 ms median handshake and 736 MiB/s record encryption; ASan/UBSan CI on Linux and macOS, 12 libFuzzer targets, nightly mutation testing (299 must-kill mutations).
- Built mldsa-authd, a login service on it: login codes, device enrolment, revocation, key rotation, recovery codes; ProVerif model of the handshake.

TideFit | TypeScript, Next.js 16, React 19, Supabase | github.com/UtkarshMitta/tide-fit 2026
App by Utkarsh Mittal; the audit, upgrade, tests, CI and launch are mine.
- Audited all 45 source files of a Next.js trip planner: 15 vulnerabilities confirmed on the running app (OAuth callbacks accepting any authorization code, a row-level security policy exposing every user), 13 fixed.
- Upgraded Next 14 to 16 and React 18 to 19 across 16 merged pull requests; npm audit 6 vulnerabilities to 0, tests 0 to 45, CI (typecheck, lint, tests) deploying to Vercel.
- Launched it to production at tide-fit.vercel.app: private file storage, an edge rate limit verified at 20 requests then 429, Strava and Google sign-in verified live.

One-time-pad allocation (team of 3) | Python, pytest 2025
github.com/happyc0der/cryptography-project-1
Team of three; the chunk-reserve protocol, proofs, simulator, tests, CI and report are my 2026 rewrite.
- Designed a wait-free one-time-pad allocation protocol for m broadcasting parties with d messages in flight, with a proof of no pad reuse and an exact worst-case waste bound.
- Cut worst-case waste at m=5, d=5, n=5,000 from 4,000 pads (static partition) to 36, with zero blocked sends across the adversarial grid.
- Found two defects in the course handout's own two-party protocol, each pinned by a test; 310 tests, and CI reruns the evaluation and diffs the results.

Hawk signature study | Python, NumPy, SymPy | github.com/happyc0der/hawk-signature-study 2026
- Studied Hawk, the lattice signature scheme withdrawn from NIST standardisation in July 2026: a 13-page report, a talk, and a Flask keygen/sign/verify API with a demo page.
- Made six correctness fixes to the vendored hawk-py reference implementation, with regression tests; one sped up Hawk-256 verification about 300x (6.04 s to 0.02 s).
- Documented the attack that broke it beside the unedited report: key recovery reduces to a lattice problem in half the dimension, cutting Hawk-512 from 2^150 to 2^108.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

Security: applied cryptography, post-quantum cryptography (ML-DSA, ML-KEM), X25519, ChaCha20-Poly1305, HKDF, network security (TLS), OAuth 2.1 (PKCE, JWT), libsodium, liboqs, fuzzing (libFuzzer), mutation testing, ASan/UBSan, formal verification (ProVerif), security auditing and vulnerability assessment, threat modeling

Systems: CMake, gRPC, PostgreSQL, SQLite, WebAssembly (sandboxed mods), Android (Jetpack Compose, Room), Next.js, React, FastAPI, Node.js, REST APIs, MCP

Testing: pytest, Catch2, vitest, Robolectric, libFuzzer, ASan/UBSan, static analysis (clang-tidy, ruff, ESLint), ProVerif, unit and integration tests, CI

LEADERSHIP AND ACTIVITIES

- Open source: ephys-mcp on PyPI and the MCP Registry (24 tools, 52 tests); six fixes with regression tests to hawk-py
- Treasurer, OWASP Student Chapter, IIIT Delhi
- Attended the first International Quantum Communication Conclave, New Delhi, 2023